Numerical Methods

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Locating roots by a change of sign**

Show that $f(x) = x^3 - 4x - 2$ has a root between $x = 2$ and $x = 3$.

$f(2) = 8 - 8 - 2 = -2$. $f(3) = 27 - 12 - 2 = 13$. Since $f(2) < 0$ and $f(3) > 0$, the sign change confirms a root lies in $(2, 3)$.

2 **[Foundational]** • **Target: Interval bisection**

Using $f(x) = x^3 - x - 1$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval containing the root.

$f(1.5) = 3.375 - 1.5 - 1 = 0.875 > 0$. Since $f(1) < 0$ and $f(1.5) > 0$, the root lies in $(1, 1.5)$.

3 **[Foundational]** • **Target: Interval bisection**

Continuing from the previous problem, evaluate $f(1.25)$ and give a narrower interval for the root.

$f(1.25) = 1.953125 - 1.25 - 1 = -0.296875 < 0$. Since $f(1.25) < 0$ and $f(1.5) > 0$, the root lies in $(1.25, 1.5)$.

4 **[Foundational]** • **Target: Interval bisection**

Using $f(x) = \ln x - 2 + x$, which changes sign between $x = 1$ and $x = 2$, evaluate $f(1.5)$ and give a narrower interval for the root.

$f(1.5) = \ln 1.5 - 2 + 1.5 = 0.405465 - 0.5 = -0.094535 < 0$. Since $f(1.5) < 0$ and $f(2) > 0$, the root lies in $(1.5, 2)$.

5 **[Foundational]** • **Target: Fixed-point iteration**

The equation $x^3 - 4x - 2 = 0$ rearranges to $x = \sqrt[3]{4x + 2}$. Using $x_0 = 2$, find x_1, x_2, and x_3, to four decimal places.

$x_1 = \sqrt[3]{4(2) + 2} = \sqrt[3]{10} \approx 2.1544$. $x_2 = \sqrt[3]{4(2.1544) + 2} = \sqrt[3]{10.6176} \approx 2.1978$.
$x_3 = \sqrt[3]{4(2.1978) + 2} = \sqrt[3]{10.7912} \approx 2.2098$ (converging toward the root ≈ 2.214).

6 **[Foundational]** • **Target: Convergence and divergence of iterations**

Explain why the iteration $x_{n+1} = x_n^3 - 3$ (i.e. $g(x) = x^3 - 3$), starting from $x_0 = 1.5$, would fail to converge to a root of $x^3 - x - 3 = 0$ near $x = 1.5$.

$g'(x) = 3x^2$. At $x = 1.5$: $g'(1.5) = 3(2.25) = 6.75$. Since $|g'(1.5)| = 6.75 > 1$, the iteration does not satisfy the convergence condition $|g'(x)| < 1$ near this point, so it diverges rather than converging to the root.

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7 [Foundational] • Target: Convergence and divergence of iterations

Using $g(x) = 5 - \frac{2}{x}$ from the earlier iteration, find $g'(x)$ and evaluate $|g'(x)|$ at $x = 4.56$ (near the root), to explain why the iteration converges there.

$g(x) = 5 - 2x^{-1}$, so $g'(x) = \frac{2}{x^2}$. At $x = 4.56$: $g'(4.56) = \frac{2}{20.7936} \approx 0.0962$. Since $|g'(4.56)| < 1$, the iteration converges near this root.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^2 - 3x - 1 = 0$ into the form $x = g(x)$ by making the linear term the subject.

From $3x = x^2 - 1$: $x = \frac{x^2 - 1}{3}$.

9 [Intermediate] • Target: Rearranging equations for iteration

Rearrange $x^3 + x - 1 = 0$ into the form $x = \frac{1}{x^2 + 1}$, and verify algebraically that this is equivalent to the original equation.

From $x^3 + x - 1 = 0$: $x(x^2 + 1) = 1$, so $x = \frac{1}{x^2 + 1}$. Verifying: multiplying both sides by $(x^2 + 1)$ gives $x(x^2 + 1) = 1$, i.e. $x^3 + x = 1$, i.e. $x^3 + x - 1 = 0$, which matches the original equation exactly.

10 [Intermediate] • Target: The Newton-Raphson method

Use one iteration of the Newton-Raphson method, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 4x - 2$, to four decimal places.

$f'(x) = 3x^2 - 4$. $f(2) = 8 - 8 - 2 = -2$. $f'(2) = 12 - 4 = 8$. $x_1 = 2 - \frac{-2}{8} = 2.25$.

11 [Intermediate] • Target: The Newton-Raphson method

Continuing from $x_1 = 2.25$ in the previous problem, find x_2, to four decimal places.

$f(2.25) = 11.390625 - 9 - 2 = 0.390625$. $f'(2.25) = 3(5.0625) - 4 = 11.1875$.
$x_2 = 2.25 - \frac{0.390625}{11.1875} \approx 2.25 - 0.0349 \approx 2.2151$.

12 [Intermediate] • Target: The Newton-Raphson method

Use the Newton-Raphson method with $x_0 = 2.5$ to find x_1 for $f(x) = x^2 - 7$ (approximating $\sqrt{7}$), to four decimal places.

$f'(x) = 2x$. $f(2.5) = 6.25 - 7 = -0.75$. $f'(2.5) = 5$. $x_1 = 2.5 - \frac{-0.75}{5} = 2.5 + 0.15 = 2.65$.

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13 [Intermediate] • Target: The Newton-Raphson method

Use the Newton-Raphson method with $x_0 = 2$ to find x_1 for $f(x) = x^3 - 2x - 5$, to four decimal places.

$f'(x) = 3x^2 - 2$. $f(2) = 8 - 4 - 5 = -1$. $f'(2) = 12 - 2 = 10$. $x_1 = 2 - \frac{-1}{10} = 2.1$.

14 [Intermediate] • Target: Further Newton-Raphson: rapid convergence and failure cases

Using $x_1 = 2.65$ from the previous problem for $f(x) = x^2 - 7$, perform a second iteration to find x_2, to four decimal places, and compare with $\sqrt{7} \approx 2.6458$.

$f(2.65) = 7.0225 - 7 = 0.0225$. $f'(2.65) = 5.3$. $x_2 = 2.65 - \frac{0.0225}{5.3} \approx 2.65 - 0.00425 \approx 2.6458$, matching $\sqrt{7}$ to four decimal places after just two iterations.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 11.01 Differential Equations

Explain why $x_0 = 1$ is an unsuitable starting point for the Newton-Raphson method applied to $f(x) = x^3 - 3x + 1$.

$f'(x) = 3x^2 - 3$. At $x_0 = 1$: $f'(1) = 3 - 3 = 0$. Since $f'(1) = 0$, the tangent line at $x_0 = 1$ is horizontal and never meets the x-axis, so the formula $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ involves division by zero — the method fails at this starting point.

16 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 12.01 Poisson Distribution

Continuing from $x_1 = 2.1$ for $f(x) = x^3 - 2x - 5$, perform a second Newton-Raphson iteration to find x_2, to four decimal places.

$f(2.1) = 9.261 - 4.2 - 5 = 0.061$. $f'(2.1) = 3(4.41) - 2 = 11.23$.
$x_2 = 2.1 - \frac{0.061}{11.23} \approx 2.1 - 0.00543 \approx 2.0946$.

17 [Challenging] • Target: Error bounds from bisection • Cross-Lesson: 13.01 Continuous Random Variables

A root of $f(x) = x^3 - 4x - 2$ lies in $(2.21, 2.22)$, since $f(2.21) \approx -0.046$ and $f(2.22) \approx 0.061$. State the midpoint estimate and the maximum possible error.

Midpoint: $\frac{2.21 + 2.22}{2} = 2.215$. Maximum possible error: half the interval width, $\frac{2.22 - 2.21}{2} = 0.005$.

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18 [Challenging] • Target: Error bounds from bisection • Cross-Lesson: 14.01 Sampling & Estimation

Given that the root of $f(x) = x^3 - 4x - 2$ is trapped in the interval $(2.2141, 2.2144)$ after further bisection, state the root correct to three decimal places, justifying your answer.

Every value in the interval $(2.2141, 2.2144)$ rounds to 2.214 at three decimal places (since both endpoints lie strictly between 2.2135 and 2.2145). The root is therefore 2.214 to three decimal places.

19 [Challenging] • Target: A well-designed word problem: designing a support strut • Cross-Lesson: 15.01 Hypothesis Testing

This question has four parts. An engineer models the cross-sectional profile of a support strut by $f(x) = x^3 - 6x^2 + 9x - 2$ (metres) and needs to find where it crosses the x-axis. (a) Show by a change of sign that a root lies between $x = 0$ and $x = 1$. (b) Use the Newton-Raphson method with $x_0 = 0.5$ to find x_1, to four decimal places. (c) Perform a second iteration to find x_2, to four decimal places. (d) Perform a third iteration to find x_3, and state the root to three decimal places.

$f'(x) = 3x^2 - 12x + 9$. (a) $f(0) = -2$ and $f(1) = 1 - 6 + 9 - 2 = 2$; the sign change confirms a root in $(0, 1)$. (b) $f(0.5) = 0.125 - 1.5 + 4.5 - 2 = 1.125$; $f'(0.5) = 0.75 - 6 + 9 = 3.75$; $x_1 = 0.5 - \frac{1.125}{3.75} = 0.2$. (c) $f(0.2) = 0.008 - 0.24 + 1.8 - 2 = -0.432$; $f'(0.2) = 0.12 - 2.4 + 9 = 6.72$; $x_2 = 0.2 - \frac{-0.432}{6.72} \approx 0.2643$. (d) $f(0.2643) \approx -0.02196$; $f'(0.2643) \approx 6.0380$; $x_3 = 0.2643 - \frac{-0.02196}{6.0380} \approx 0.2679$. The root is 0.268 to three decimal places.

20 [Challenging] • Target: Interval bisection • Cross-Lesson: 2.01 Exponentials & Logarithms

Show that $x^3 + 2x - 6 = 0$ has a root in $[1, 2]$. Taking $x_0 = 1.5$, perform one Newton-Raphson iteration.

$f(1) = -3 < 0$, $f(2) = 6 > 0$: sign change confirms root. $f(1.5) = 1.5^3 + 3 - 6 = -0.625$. $f'(x) = 3x^2 + 2 \Rightarrow f'(1.5) = 8.75$. $x_1 = 1.5 - \frac{-0.625}{8.75} \approx 1.571$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Locating roots by a change of sign

Using the Newton-Raphson method with $f(x) = x^2 - 18$ and $x_0 = 6$, find x_1.

a. $-\frac{9}{2}$

b. $\frac{9}{2}$ ✓

c. 6

d. $\frac{11}{2}$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = \frac{x_0^2 + 18}{2x_0} = \frac{9}{2}$.

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22 [Foundational] • Target: Interval bisection

The unique real root α of the equation $x^3 + 2x - k = 0$, where k is a positive constant, is estimated using the iterative formula $x_{n+1} = \frac{k}{x_n^2 + 2}$. The sequence of iterations converges locally to α (for an initial approximation x_0 sufficiently close to α) if and only if $|g'(\alpha)| < 1$, where $g(x) = \frac{k}{x^2 + 2}$. Determine the complete set of values of k for which this iteration converges locally to α.

a. $0 < k < 4\sqrt{2}$ ✓

b. $0 < k < 2\sqrt{2}$

c. $0 < k < 3$

d. $0 < k < 12$

To determine the range of k for which the iteration converges locally, we follow these steps: 1. **Root relation:** At the fixed point $x = \alpha$, we have: $\alpha = g(\alpha) = \frac{k}{\alpha^2 + 2} \Rightarrow \alpha^3 + 2\alpha = k$. Since $k > 0$ and the function $f(x) = x^3 + 2x - k$ satisfies $f'(x) = 3x^2 + 2 > 0$ for all real x, there is a unique real root α, and $\alpha > 0$. 2. **Differentiating $g(x)$:** Using the chain rule on $g(x) = k(x^2 + 2)^{-1}$.

23 [Foundational] • Target: Fixed-point iteration

An interval of width 32 containing a root must be reduced to width at most 1 using repeated bisection. Find the minimum number of bisections required.

a. 7

b. 4

c. 6

d. **5** ✓

Halving repeatedly, after n bisections the width is $32/2^n$. The smallest n with $32/2^n \le 1$ is $n = 5$ (width becomes 1).

24 [Foundational] • Target: Convergence and divergence of iterations

An interval of width 8 containing a root is bisected 3 times. Find the width of the resulting interval.

a. 2

b. $\frac{13}{12}$

c. $\frac{8}{3}$

d. **1** ✓

Each bisection halves the width: $\frac{8}{2^3} = 1$.

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25 [Foundational] • Target: Rearranging equations for iteration

A curve passes through the point $(1, \frac{1}{2})$ and satisfies the differential equation $\frac{dy}{dx} = \frac{y^2(1-\ln x)}{x^2}$ for $x > 0$. Determine the exact coordinates of the stationary point of the curve and its nature.

a. $(e, \frac{e}{2e-1})$**, which is a local maximum ✓**

b. $(e, \frac{1}{2e-1})$, which is a local maximum

c. $(e, \frac{e}{2e+1})$, which is a local minimum

d. $(\frac{1}{e}, \frac{1}{2e+1})$, which is a local minimum

1. **Separate the variables:** $\frac{1}{y^2}\,dy = \frac{1-\ln x}{x^2}\,dx$ 2. **Integrate both sides:** The left-hand side integrates to: $\int y^{-2}\,dy = -\frac{1}{y}$ For the right-hand side, split into two integrals: $\int \frac{1-\ln x}{x^2}\,dx = \int x^{-2}\,dx - \int x^{-2}\ln x\,dx$ We have $\int x^{-2}\,dx = -\frac{1}{x}$. Using integration by parts for $\int x^{-2}\ln x\,dx$ with $u = \ln x$ and $v' = x^{-2}$.

26 [Foundational] • Target: The Newton-Raphson method

The continuous function f is defined for all $x \in \mathbb{R}$ by $f(x) = 2^x + 3x - 11$. Between which pair of consecutive integers does a root of the equation $f(x) = 0$ lie?

a. 2 **and** 3 **✓**

b. 1 and 2

c. 3 and 4

d. 0 and 1

To determine the interval containing a root of a continuous function $f(x)$, we use the change-of-sign rule (Intermediate Value Theorem). If $f(a)$ and $f(b)$ have opposite signs, then at least one real root of $f(x) = 0$ lies in the interval (a, b). Evaluate $f(x)$ at consecutive integers: - For $x = 0$: $f(0) = 2^0 + 3(0) - 11 = 1 + 0 - 11 = -100$ - For $x = 4$: $f(4) = 2^4 + 3(4) - 11 = 16 + 12 - 11 = 17 > 0$ Since $f(2) = -10$, there is a change of sign between $x = 2$ and $x = 3$. Because f is continuous.

27 [Foundational] • Target: Further Newton-Raphson: rapid convergence and failure cases

A confidence interval for the mean uses $z = 1$, $\sigma = 4$, $n = 9$. Find the margin of error $E = z\sigma/\sqrt{n}$.

a. $\frac{7}{3}$

b. $\frac{4}{3}$ **✓**

c. $\frac{17}{12}$

d. $\frac{8}{3}$

Applying Sampling & Estimation principles, $E = 1 \times 4/3 = \frac{4}{3}$. Therefore, the verified correct option is $\frac{4}{3}$.

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28 [Foundational] • Target: Error bounds from bisection

Which differential equation correctly models the statement: "The rate of increase of the number of infected people N is proportional to the product of N and the number of remaining susceptible people $(S - N)$, for fixed population S."?

a. $\frac{dm}{dt} = -km$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dT}{dt} = -k(T - T_e)$

d. $\frac{dN}{dt} = kN(S - N)$ ✓

The correct model is $\frac{dN}{dt} = kN(S - N)$, where $k > 0$ is a constant of proportionality.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: designing a support strut

A sequence is defined by $x_{n+1} = \frac{2x+6}{2}$ with $x_0 = -2$. Find x_1.

a. 2

b. -2

c. **1** ✓

d. -1

Applying the iteration 1 time(s) from $x_0 = -2$: $x_1 = 1$.

30 [Intermediate] • Target: Locating roots by a change of sign

A convergent sequence satisfies $x_{n+1} = \frac{2x+2}{4}$. Find its limit L.

a. **1** ✓

b. $\frac{1}{2}$

c. $\frac{1}{3}$

d. 2

At the limit, $4L = 2L + 2 \Rightarrow L = \frac{2}{4-(2)} = 1$.

31 [Intermediate] • Target: Interval bisection

An object cools according to Newton's Law of Cooling, $T = T_{env} + (T_0 - T_{env})e^{-kt}$, with $T_{env} = 25$ and $T_0 = 41$. Find T at the time t when $e^{-kt} = 1/4$.

a. 41

b. 21

c. **29** ✓

d. 30

$T = 25 + (41 - 25) \times \frac{1}{4} = 25 + 16/4 = 25 + 4 = 29$.

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32 [Intermediate] • Target: Fixed-point iteration

The function f is defined on the open interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ by $f(x) = \frac{2-\sin x}{\cos x} + \frac{\cos x}{2-\sin x}$. Determine the exact minimum value of $f(x)$.

a. $\frac{4\sqrt{3}}{3}$ ✓

b. 2

c. $\frac{5}{2}$

d. $\frac{3\sqrt{3}}{2}$

Let $u(x) = \frac{2-\sin x}{\cos x}$. For $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we have $\cos x > 0$ and $2 - \sin x > 0$, so $u(x) > 0$. The given expression is of the form $g(u) = u + \frac{1}{u}$. **Step 1: Check feasibility of the standard AM-GM minimum** By the AM-GM inequality, $u + \frac{1}{u} \geq 2$, with equality if and only if $u = 1$. Setting $u = 1$: $\frac{2-\sin x}{\cos x} = 1 \Rightarrow \sin x + \cos x = 2$. Using the harmonic R-addition form.

33 [Intermediate] • Target: Convergence and divergence of iterations

The curve C has equation $y = \frac{\ln x}{x^2}, \quad x > 0$. The point M is the stationary point of C, and the line L is the tangent to C at M. The finite region R is bounded by the curve C, the tangent line L, and the vertical line $x = 1$. Find the exact area of R.

a. $\frac{4\sqrt{e}-2e-1}{2e}$ ✓

b. $\frac{4\sqrt{e}-2e+1}{2e}$

c. $\frac{2e-3\sqrt{e}}{2e}$

d. $\frac{2\sqrt{e}-2e-1}{2e}$

1. **Find the stationary point M:** Using the quotient rule to differentiate $y = \frac{\ln x}{x^2}$: $\frac{dy}{dx} = \frac{\frac{1}{x} \cdot x^2 - (\ln x)(2x)}{x^4} = \frac{x(1-2\ln x)}{x^4} = \frac{1-2\ln x}{x^3}$. Setting $\frac{dy}{dx} = 0$: $1 - 2\ln x = 0 \Rightarrow \ln x = \frac{1}{2} \Rightarrow x = e^{1/2} = \sqrt{e}$. The y -coordinate at M is: $y = \frac{\ln(\sqrt{e})}{(\sqrt{e})^2} = \frac{\frac{1}{2}}{e} = \frac{1}{2e}$. Since M is a stationary point, the tangent line L is horizontal wit.

34 [Intermediate] • Target: Rearranging equations for iteration

Determine the complete set of real values of the constant k for which the equation $e^x - kx^2 = 0$ has exactly three distinct real roots.

a. $k > \frac{e^2}{4}$ ✓

b. $k \geq \frac{e^2}{4}$

c. $0 < k < \frac{e^2}{4}$

d. $k > \frac{e}{2}$

First, note that $x = 0$ is not a solution to $e^x - kx^2 = 0$ since $e^0 = 1eq0$. Therefore, we can rewrite the equation as: $k = \frac{e^x}{x^2}$ Let $f(x) = \frac{e^x}{x^2}$ defined for all $xeq0$. The number of distinct real roots corresponds to the number of points of intersection between the curve $y = f(x)$ and the horizontal line $y = k$. Differentiating $f(x)$ using the quotient rule: $f'(x) = \frac{e^x \cdot x^2 - e^x \cdot 2x}{x^4} = \frac{e^x x(x-2)}{x^4} = \frac{e^x(x-2)}{x^3}$ Setting.

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35 [Intermediate] • Target: The Newton-Raphson method

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. **44** ✓

b. 43

c. 11

d. 22

We solve the problem in two stages: **Step 1: Determine σ** Under H_0, the distribution of the sample mean is $\bar{X} \sim N\left(20.0, \frac{\sigma^2}{25}\right)$. The critical value for a upper one-tailed test at the 5% level corresponds to $z = 1.645$: $\frac{20.658 - 20.0}{\frac{\sigma}{\sqrt{25}}} = 1.645$ $\frac{0.658}{\frac{\sigma}{5}} = 1.645 \Rightarrow \frac{3.29}{\sigma} = 1.645 \Rightarrow \sigma = \frac{3.29}{1.645} = 2.0$ mm **Step 2: Determine the minimum sample size m** For.

36 [Intermediate] • Target: Further Newton-Raphson: rapid convergence and failure cases

A curve satisfies the differential equation $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{k - y}{2x + 1}$ for $x > -\frac{1}{2}$, where k is a real constant. The curve passes through the points $(0, 8)$ and $(4, 2)$. Find the value of k.

a. -1 ✓

b. 11

c. $\frac{5}{4}$

d. 5

To find the constant k, we solve the differential equation by separation of variables: $\int \frac{1}{k - y}\,\mathrm{d}y = \int \frac{1}{2x + 1}\,\mathrm{d}x$ Integrating both sides (paying close attention to the chain rule): $-\ln|k - y| = \frac{1}{2}\ln(2x + 1) + C$ Multiplying by -1 and exponentiating: $\ln|k - y| = -\frac{1}{2}\ln(2x + 1) - C = \ln\left((2x + 1)^{-1/2}\right) - C$
$|k - y| = \frac{A}{\sqrt{2x + 1}}$ where $A = \mathrm{e}^{-C} > 0$ Removing the absolute value gives th.

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37 [Intermediate] • Target: Error bounds from bisection

A curve has equation $y = x^2e^{3x}$. Find the gradient of the curve at the point where $x = 1$.

a. $5e^3$ ✓

b. $6e^3$

c. $3e^3$

d. $2e^3$

To find the gradient of the curve, differentiate $y = x^2e^{3x}$ with respect to x using the product rule: $\frac{d}{dx}[uv] = u\frac{dv}{dx} + v\frac{du}{dx}$ Let $u = x^2$ and $v = e^{3x}$. Then $\frac{du}{dx} = 2x$ and, by the chain rule, $\frac{dv}{dx} = 3e^{3x}$. Applying the product rule: $\frac{dy}{dx} = x^2(3e^{3x}) + 2x(e^{3x}) = (3x^2 + 2x)e^{3x}$ Substitute $x = 1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: designing a support strut • Cross-Lesson: 14.01 Sampling & Estimation

Relative to a fixed origin O, the position vectors of the points A and B are given by $\vec{OA} = \mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} - 8\mathbf{k}$. Find the unit vector in the direction of $\vec{AB}$.

a. $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$ ✓

b. $\frac{1}{13}(-3\mathbf{i} - 4\mathbf{j} + 12\mathbf{k})$

c. $\frac{1}{169}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

d. $\frac{1}{19}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$

First, find the displacement vector $\vec{AB}$: $\vec{AB} = \vec{OB} - \vec{OA} = (4-1)\mathbf{i} + (2-(-2))\mathbf{j} + (-8-4)\mathbf{k} = 3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k}$. Next, calculate the magnitude of $\vec{AB}$: $|\vec{AB}| = \sqrt{3^2 + 4^2 + (-12)^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = 13$. The unit vector in the direction of $\vec{AB}$ is given by $\frac{\vec{AB}}{|\vec{AB}|}$: $\frac{1}{13}(3\mathbf{i} + 4\mathbf{j} - 12\mathbf{k})$. Common misconceptions: - Reversing the subtraction (cal.

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39 [Challenging] • Target: Locating roots by a change of sign • Cross-Lesson: 15.01 Hypothesis Testing

Relative to a fixed origin O, the points A and B have position vectors $\overrightarrow{OA} = \mathbf{i} + 5\mathbf{j} - 2\mathbf{k}$ and $\overrightarrow{OB} = 3\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. Find the length of the vector $\overrightarrow{AB}$.

a. 7 ✓

b. $\sqrt{31}$

c. $\sqrt{69}$

d. 49

To find the vector $\overrightarrow{AB}$, subtract the position vector of A from that of B: $\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (3-1)\mathbf{i} + (2-5)\mathbf{j} + (4-(-2))\mathbf{k} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$. The magnitude (length) of $\overrightarrow{AB}$ is: $|\overrightarrow{AB}| = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Common misconceptions: - Incorrectly taking $(-3)^2 = -9$ leads to $\sqrt{4 - 9 + 36} = \sqrt{31}$. - Adding th.

40 [Challenging] • Target: Interval bisection • Cross-Lesson: 2.01 Exponentials & Logarithms

Find the particular solution of the differential equation $\frac{dy}{dx} = 2xe^{-y}$ given that $y = 0$ when $x = 0$.

a. $y = \ln(x^2 + 1)$ ✓

b. $y = \ln(x^2)$

c. $y = \ln(2x^2 + 1)$

d. $y = x^2$

To solve the differential equation, we use separation of variables: 1. **Separate the variables**: $e^y\,dy = 2x\,dx$ 2. **Integrate both sides**: $\int e^y\,dy = \int 2x\,dx$ $e^y = x^2 + C$ 3. **Apply the initial condition** $y = 0$ when $x = 0$: $e^0 = 0^2 + C \Rightarrow 1 = C$ *(A frequent error is assuming $C = 0$ because the boundary values are zero, which leads to the incorrect solution $y = \ln(x^2)$.)* 4. **Express y explicitly in terms of x**.

41 [Challenging] • Target: Fixed-point iteration • Cross-Lesson: 3.01 Advanced Trigonometry

The continuous random variable X has probability density function $f(x)$ defined by \[f(x) = \begin{cases} kx & 0 \le x 1 \mid X < 2.5) \).

a. $\frac{9}{11}$ ✓

b. $\frac{10}{11}$

c. $\frac{3}{4}$

d. $\frac{9}{10}$

Step 1: Determine the constant k using the total probability property $\int_{-\infty}^{\infty} f(x)\,dx = 1$. $\int_0^2 kx\,dx + \int_2^3 k(6-2x)\,dx = 1$ Evaluating each integral: $\int_0^2 kx\,dx = k\left[\frac{x^2}{2}\right]_0^2 = 2k$ $\int_2^3 k(6-2x)\,dx = k[6x - x^2]_2^3 = k((18-9) - (12-4)) = k(9-8) = k$ Thus, $2k + k = 3k = 1 \Rightarrow k = \frac{1}{3}$. Step 2: Find P($X1$ | $X1$) instead of.

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42 [Challenging] • Target: Convergence and divergence of iterations • Cross-Lesson: 4.01 Differentiation

A curve satisfies $x^2 + y^2 = 25$. Find $\frac{dy}{dx}$ at the point $(3, -4)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$ ✓

c. $\frac{4}{3}$

d. $-\frac{4}{3}$

Differentiating implicitly: $2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$. At $(3, -4)$: $\frac{3}{4}$.

43 [Challenging] • Target: Rearranging equations for iteration • Cross-Lesson: 5.01 Integration

Which differential equation correctly models the statement: "The rate of decrease of the temperature T of an object is proportional to the difference between T and the surrounding temperature T_e."?

a. $\frac{dP}{dt} = kP$

b. $\frac{dT}{dt} = -k(T - T_e)$ ✓

c. $\frac{dN}{dt} = kN(S - N)$

d. $\frac{dm}{dt} = -km$

The correct model is $\frac{dT}{dt} = -k(T - T_e)$, where $k > 0$ is a constant of proportionality.

44 [Challenging] • Target: The Newton-Raphson method • Cross-Lesson: 6.01 Sequences & Series

Which differential equation correctly models the statement: "The rate of increase of the volume V of a balloon is proportional to the square of its radius r."?

a. $\frac{dT}{dt} = -k(T - T_e)$

b. $\frac{dm}{dt} = -km$

c. $\frac{dV}{dt} = kr^2$ ✓

d. $\frac{dP}{dt} = kP$

The correct model is $\frac{dV}{dt} = kr^2$, where $k > 0$ is a constant of proportionality.

45 [Challenging] • Target: Further Newton-Raphson: rapid convergence and failure cases • Cross-Lesson: 7.01 Vectors

Which differential equation correctly models the statement: "The rate of decrease of a radioactive mass m is proportional to m."?

a. $\frac{dN}{dt} = kN(S - N)$

b. $\frac{dV}{dt} = kr^2$

c. $\frac{dP}{dt} = kP$

d. $\frac{dm}{dt} = -km$ ✓

The correct model is $\frac{dm}{dt} = -km$, where $k > 0$ is a constant of proportionality.

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